Diogo Gil

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CONTACT

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EDUCATION

MSc in Finance
Nova School of Business & Economics
2026 - 2027 | In progress

MSc in Applied Mathematics
FCUL - Ciências de Lisboa
2025 (Transitioned to Finance)
Coursework: Stats, Probability, Data Science, Advanced Analysis, OR Models.

BSc in Physics
FCUL - Ciências de Lisboa
2022 - 2025 | Minor in Comp. Eng.

TECHNICAL SKILLS

PROGRAMMING
Python (NumPy, Pandas, SciPy, scikit-learn), SQL, Java

QUANTITATIVE MODELLING
Monte Carlo simulation (10k–100k runs), VaR/CVaR, Kupiec backtesting, stress testing

QUANTITATIVE METHODS
Time Series (ARIMA, GARCH), regression modelling, volatility modelling, numerical optimization

RISK & FINANCE
Portfolio modelling, risk metrics, scenario analysis

ANALYTICS TOOLS
Power BI, Advanced Excel, LaTeX, Git, VS Code

SELECTED SPECIALIZED CERTIFICATIONS

- YALE UNIVERSITY — Financial Markets
- COLUMBIA UNIVERSITY — Financial Eng. & Risk
- IESE BUSINESS SCHOOL — Corporate Finance
- CAMBRIDGE — C1 Advanced

Online Portfolio:
diogogil2727.github.io/Portfolio

Diogo Gil

FINANCE STUDENT (NOVA SBE) & QUANTITATIVE RESEARCH ASPIRANT

PROFESSIONAL PROFILE

Quantitative-minded Finance student with a strong background in Physics, Applied Mathematics and Computer Science. Completed advanced coursework in probability, statistics, optimization, multivariate analysis and data science before transitioning into Finance to apply analytical reasoning to financial markets, risk modelling and data-driven decision-making. Experienced in Python-based modelling, risk analysis and automation. Interested in quantitative research, systematic strategies and financial engineering.

QUANTITATIVE COURSEWORK

Linear Algebra & Analytical Geometry — vector spaces, eigenvalues, diagonalization
Calculus I, II, III — multivariate calculus, series, differential equations
Probability & Statistics — distributions, inference, hypothesis testing
Numerical Methods — ODE/PDE solvers, interpolation, root-finding
Analytical Mechanics — variational calculus, Hamiltonian systems
Methods of Mathematical Physics — special functions, Fourier/Laplace transforms
Statistical Physics — stochastic processes, Markov models
Quantum Mechanics I & II — linear algebra, operators, eigenvalue problems
Computational Physics — numerical simulation, Monte Carlo methods
Mathematical Computing — algorithms, numerical optimization
Probability & Statistics (MSc) — advanced inference, multivariate statistics
Data Science & Practical Statistics (MSc) — regression, ML fundamentals
Operations Research Models (MSc) — optimization, linear/quadratic programming

PROFESSIONAL EXPERIENCE

BNP Paribas — Rising Talents — Risk Trainee
Oct 2025 - Sep 2026

- Selected for BNP Paribas' competitive Rising Talents program, aimed at developing future leaders in risk management.
- **Migrated legacy SAS risk processes to Python** as part of the transition away from SAS, translating existing analytical workflows into scalable and maintainable Python pipelines.
- Automated monthly reporting processes, **reducing manual work by 20 hours**.
- Contributed to **portfolio monitoring, risk analysis, and stress testing** in consumer finance.
- Collaborated with cross-functional teams, gaining hands-on exposure to operational risk frameworks.

OTHER PROFESSIONAL EXPERIENCE

Tutored Physics and Mathematics at secondary and university levels (2 years). Waiter at two different restaurants during the summers of 2022 and 2023, developing client relations and high-tempo teamwork.

QUANTITATIVE & TECHNICAL PROJECTS

Quantitative Market Risk Dashboard
Independent Project

Built a Python/Streamlit dashboard for real-time market risk monitoring. Implemented **Historical, Parametric and Monte Carlo VaR** (95%), running **10k–100k simulations**. Computed **CVaR** and tail-risk metrics using distribution-based and simulation-based methods. Performed **Kupiec backtesting** (63 breaches, LR = 0.0007, p = 0.9794) to validate model accuracy. Designed **stress testing scenarios** using real historical events (COVID crash, 2008 crisis, Ukraine invasion). Automated data ingestion via **yfinance**, with full documentation and reproducible pipelines.

Early Universe Galaxy Spectral Analysis
JWST Research Project

Analyzed high-redshift galaxy spectra from the James Webb Space Telescope using advanced spectral fitting, applying statistical regressions to infer gas environments and stellar properties.